

Statistical Modeling

DATA.STAT.740

Lecture 8

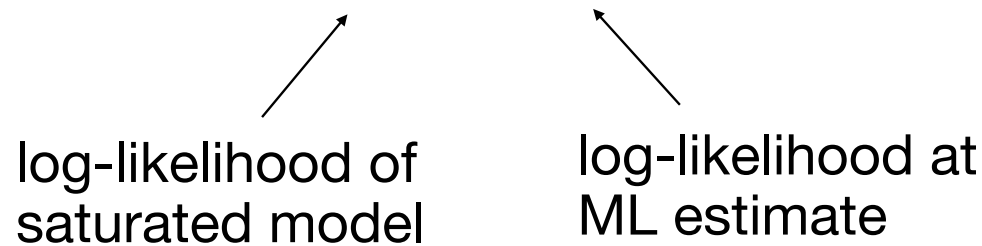
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Deviance details via Bernoulli example

Deviance is used in various places in GLM estimation

$$D(y, \hat{\mu}) = 2 [l(y, y) - l(y, \hat{\mu})]$$

log-likelihood of
saturated model



log-likelihood at
ML estimate

Let's look at the Bernoulli model as an example

$$l(\beta) = l(y_i, \mu_i) = \sum_{i=1}^N (y_i \log(\mu_i) + (1 - y_i) \log(1 - p_i))$$

Deviance details via Bernoulli example

Start with the log likelihood

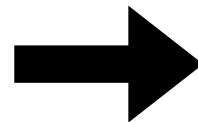
$$l(\beta) = l(y_i, \mu_i) = \sum_{i=1}^N (y_i \log(\mu_i) + (1 - y_i) \log(1 - \mu_i))$$

The saturated model fits all the observations perfectly. In Bernoulli case, plug in observations and calculate $l(y, y)$:

$$l(y, y) = \sum_{i=1}^N y_i \log(y_i) + (1 - y_i) \log(1 - y_i), \quad (0 \cdot \log 0 = 0)$$

$$y_i = 1 : 1 \cdot \log(1) + 0 \cdot \log(0) = 0$$

$$y_i = 0 : 0 \cdot \log(0) + 1 \cdot \log(1 - 0) = 0$$



Saturated model: $l(y, y) = 0$

Deviance: $D(y, \hat{\mu}) = -l(y, \hat{\mu})$

Binomial regression

Outcome is y_i successes out of m_i possibilities

Logistic regression is a special case of Binomial regression $m_i = 1, \quad i = 1, \dots, N$

Binomial regression is also called logistic regression (when logit-link is used!)

In Binomial regression, the observations have a Binomial distribution

$$\mathbf{y}_i \sim \text{Binom}(\mu_i, p_i) = \binom{m_i}{y_i} p_i^{y_i} (1 - p_i)^{m_i - y_i}$$

$$\text{E}(\mathbf{y}_i) = \mu_i = m_i p_i$$

$$\text{Var}(\mathbf{y}_i) = m_i p_i (1 - p_i)$$

Binomial regression

Systematic component $\eta_i = x_i^T \beta$

Logit link $g(p_i) = \text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right)$
$$p_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

Mean model $\mu_i = m_i p_i$

Variance function $V(\mu_i) = \frac{\mu_i}{m_i} \left(1 - \frac{\mu_i}{m_i} \right) m_i = \mu_i \left(1 - \frac{\mu_i}{m_i} \right)$

log-likelihood

$$l(\beta) = \sum_{i=1}^N (y_i \log(p_i) + (m_i - y_i) \log(1 - p_i))$$

Score function for the canonical link

log-likelihood $l(\beta) = \sum_{i=1}^N (y_i \log(p_i) + (m_i - y_i) \log(1 - p_i))$

For canonical link $\frac{\partial p_i}{\partial \eta_i} = p_i(1 - p_i)$ (from last week $\frac{\partial \mu}{\partial \eta} = \frac{\text{Var}(\mathbf{y})}{a(\phi)}$)

and the score equations are $\mathbf{u}(\beta) = \frac{\partial l}{\partial \beta} = \sum_{i=1}^N x_i(y_i - m_i p_i) = 0$

and Fisher information matrix $F_\beta = X^T W X$, $w_{ii} = m_i p_i (1 - p_i)$

Matrix $D = \text{diag} \left(\left[\frac{\partial p_1}{\partial \eta_1}, \dots, \frac{\partial p_N}{\partial \eta_N} \right] \right)$

IRLS for solving the score equations

The Fisher scoring process can be written as

$$\begin{aligned}\beta_{t+1} &= \beta_t + F_t^{-1} \mathbf{u}_t \\ F_t \beta_{t+1} &= F_t \beta_t + \mathbf{u}_t\end{aligned}$$

and setting $F_t = X^T W_t X$, $\mathbf{u}_t = X^T D_t V^{-1}(\mathbf{y} - \mu_t) = X^T W_t D_t^{-1}(\mathbf{y} - \mu_t)$
we get

$$\begin{aligned}X^T W_t X \beta_{t+1} &= X^T W_t (X \beta_t + D_t^{-1}(\mathbf{y} - \mu_t)) \\ X^T W_t X \beta_{t+1} &= X^T W_t \mathbf{z}_t\end{aligned}$$

where $\mathbf{z}_t = X \beta_t + D_t^{-1}(\mathbf{y} - \mu_t)$ is **adjusted response variable**

and $\beta_{t+1} = (X^T W_t X)^{-1} X^T W_t \mathbf{z}_t$ is **the weighted least squares estimate**

IRLS for solving the score equations

Start with β_0

Repeat until convergence

$$\mathbf{z}_t = X\beta_t + \begin{bmatrix} \frac{\mathbf{y}_1 - x_1^T \beta_t}{p_1(1-p_1)} \\ \vdots \\ \frac{\mathbf{y}_N - x_N^T \beta_t}{p_N(1-p_N)} \end{bmatrix}$$

$$\beta_{t+1} = (X^T W_t X)^{-1} X^T W_t \mathbf{z}_t$$

Multinomial regression

Bernoulli random variable has two possible outcomes

When an observation has K distinctive (categorical) outcomes, the observation can be modeled using multinomial distribution

Define indicator variables $y_{i,k} = \begin{cases} 1 & \text{if observation } i \text{ is in category } k \\ 0 & \text{otherwise} \end{cases}$

For each i , $\sum_{k=1}^K y_{i,k} = 1$

Now the indicator variables have a multinomial distribution

$$(\mathbf{y}_{i,1}, \dots, \mathbf{y}_{i,K}) \sim \text{Multinom}(1, p_{i,1}, \dots, p_{i,K}) = \prod_{k=1}^K p_{i,k}^{y_{i,k}}$$

Log-likelihood
$$l(\beta) = \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(p_{i,k})$$

Systematic component

We select one of the categories as the reference, let's say category 1

For each $k = 2, \dots, K$ we define

$$\eta_{i,k} = x_i^T \beta_k$$

Link function

Generalized logit
$$\log \left(\frac{p_{i,k}}{p_{i,1}} \right) = \eta_{i,k} = x_i^T \beta_k$$

Inverse link (***softmax form***)

$$p_{i,1} = \frac{1}{1 + \sum_{j=2}^K \exp(x_i^T \beta_j)}$$

$$p_{i,k} = \frac{\exp(x_i^T \beta_k)}{1 + \sum_{j=2}^K \exp(x_i^T \beta_j)}, \quad k = 2, \dots, K$$

Example: laser-scanning

Dataset: laserscanning.txt

Based on airborne laser scanning, a total of $N = 1795$ trees could be identified in a forest area, and measures of the heights and widths of the trees were recorded. From the identified trees, 50 were randomly selected and the species of these trees were also recorded. Can we use the heights and the widths to identify the trees?

End of Lecture 8